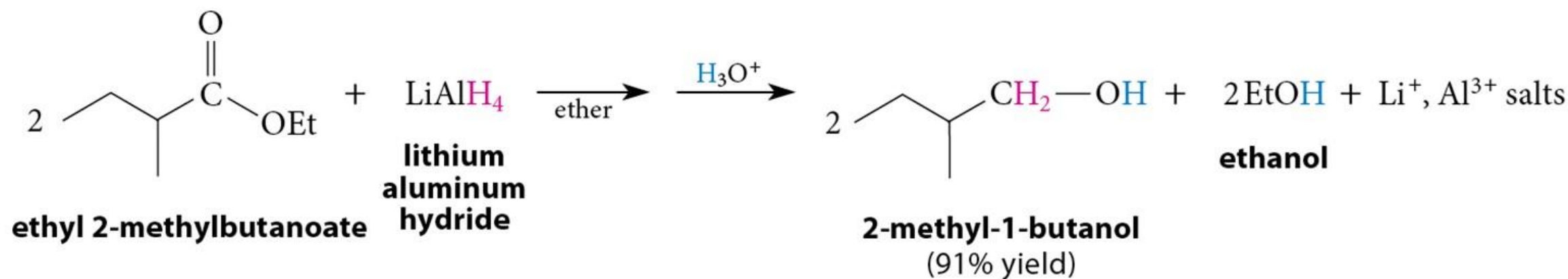


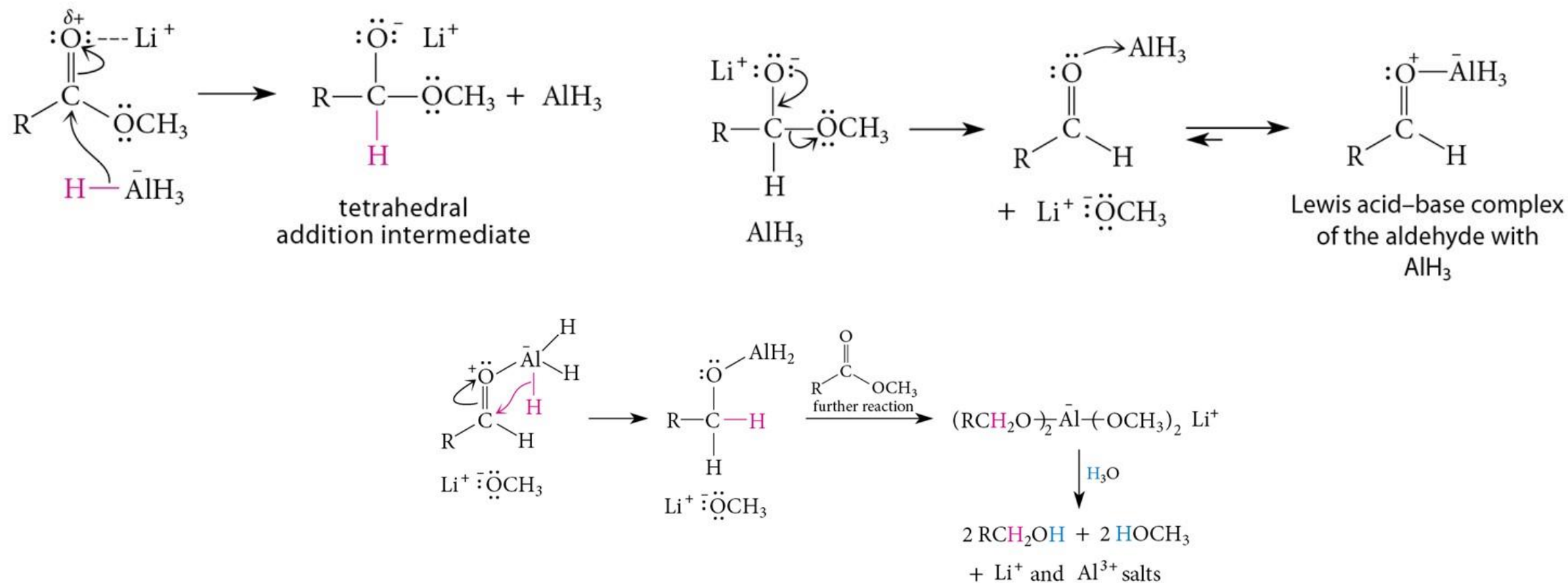
Chapter 21 (Carey Chapter 20)

The Chemistry of Carboxylic Acid Derivatives

Reduction of Esters to Primary Alcohols

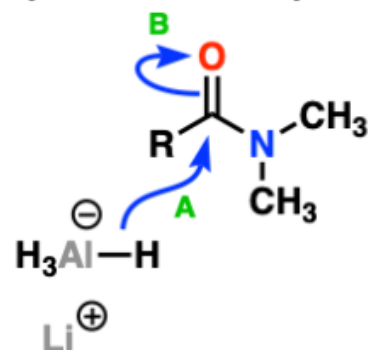


Reduction of Esters to Primary Alcohols: Mechanism

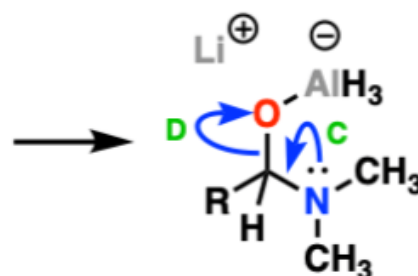


Reduction of tertiary amide

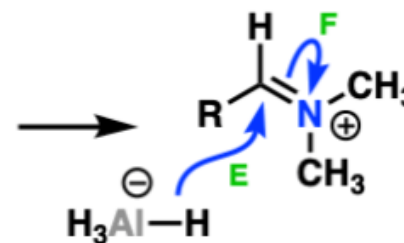
Step 1: Addition of
hydride to carbonyl



Step 2: Elimination of oxygen



Step 3: Addition of hydride to iminium

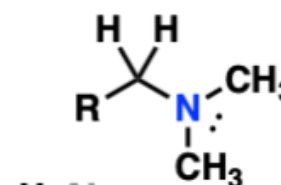


**Bonds
Formed**

A C-H
C C-N(π)
E C-H

**Bonds
Broken**

A Al-H
B C-O(π)
D C-O
E Al-H
F C-N(π)

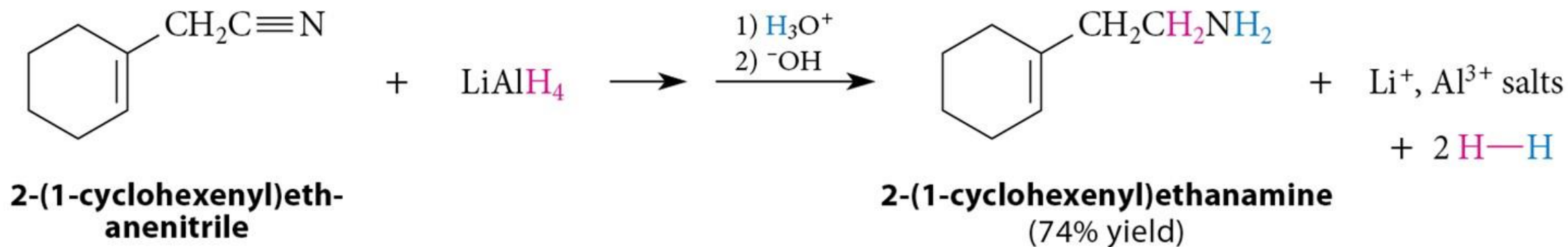


Amine

<https://www.masterorganicchemistry.com/reaction-guide/reduction-of-amides-to-amines/>

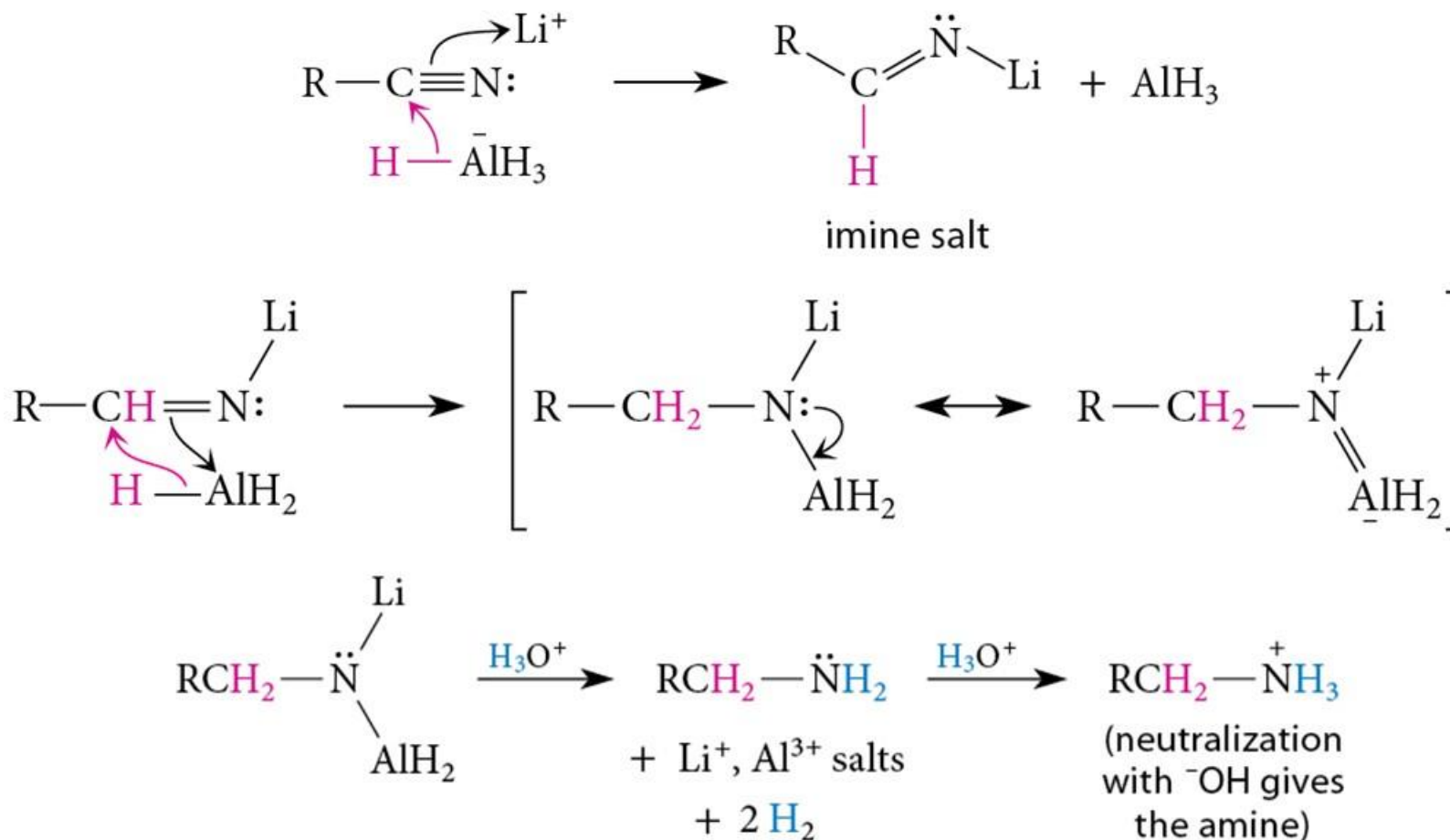
- Oxophilicity of aluminum

Reduction of Nitriles to Primary Amines: LiAlH_4



- Note that addition of base is required to isolate the neutral amine.

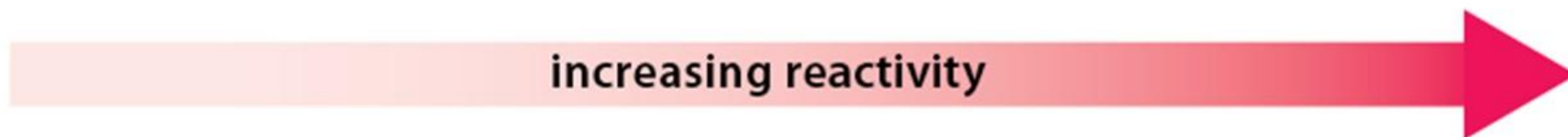
Reduction of Nitriles to Primary Amines: LiAlH_4 Mechanism



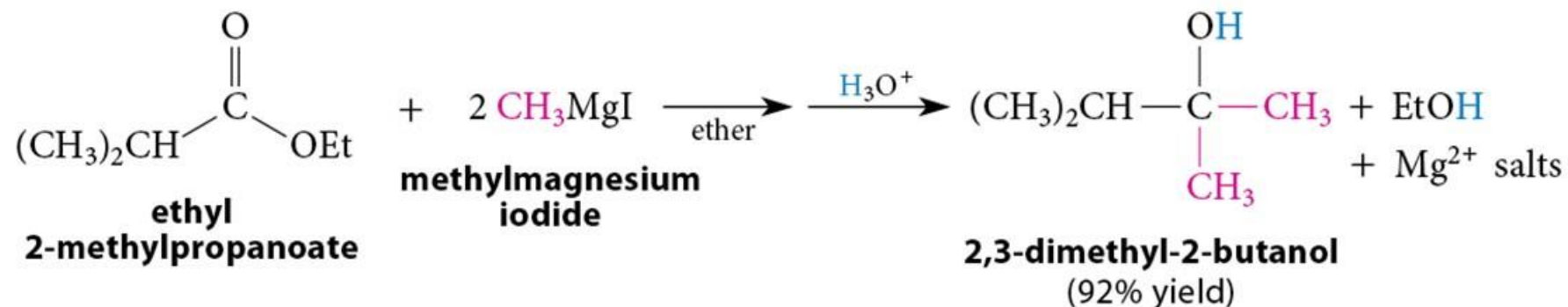
LAH Reduction and Relative Reactivities

- This reactivity order explains why the LAH reduction of an ester does not stop at the aldehyde intermediate.
- The aldehyde that is formed is *more* reactive than the ester.

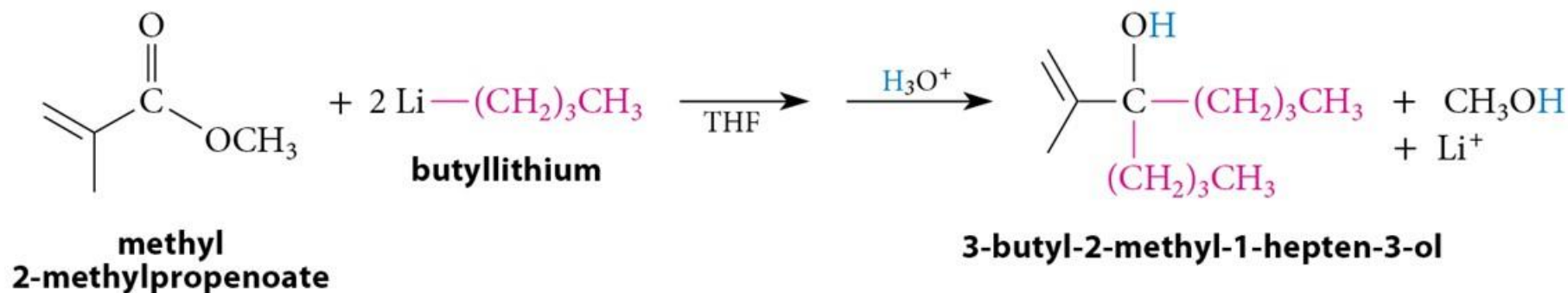
nitriles < amides < esters, acids << ketones < aldehydes < acid chlorides



Reactions of Esters with Grignard Reagents



Grignard addition happens on aldehyde, ketone, and ester (and also acid chloride and anhydrides). But not amides and nitriles.



Summary of Carboxylic Acid Derivative Preparation (1 of 2)

Synthesis of esters:

1. Acid-catalyzed esterification of RCO_2H
2. Alkylation of RCO_2H or RCO_2^-
3. Acid chlorides or anhydrides with ROH
4. Transesterification

Synthesis of acid chlorides:

- RCO_2H with SOCl_2 or PCl_5

Summary of Carboxylic Acid Derivative Preparation (2 of 2)

Synthesis of anhydrides:

1. RCO_2H with dehydrating agents
2. Acid chlorides with RCO_2^-

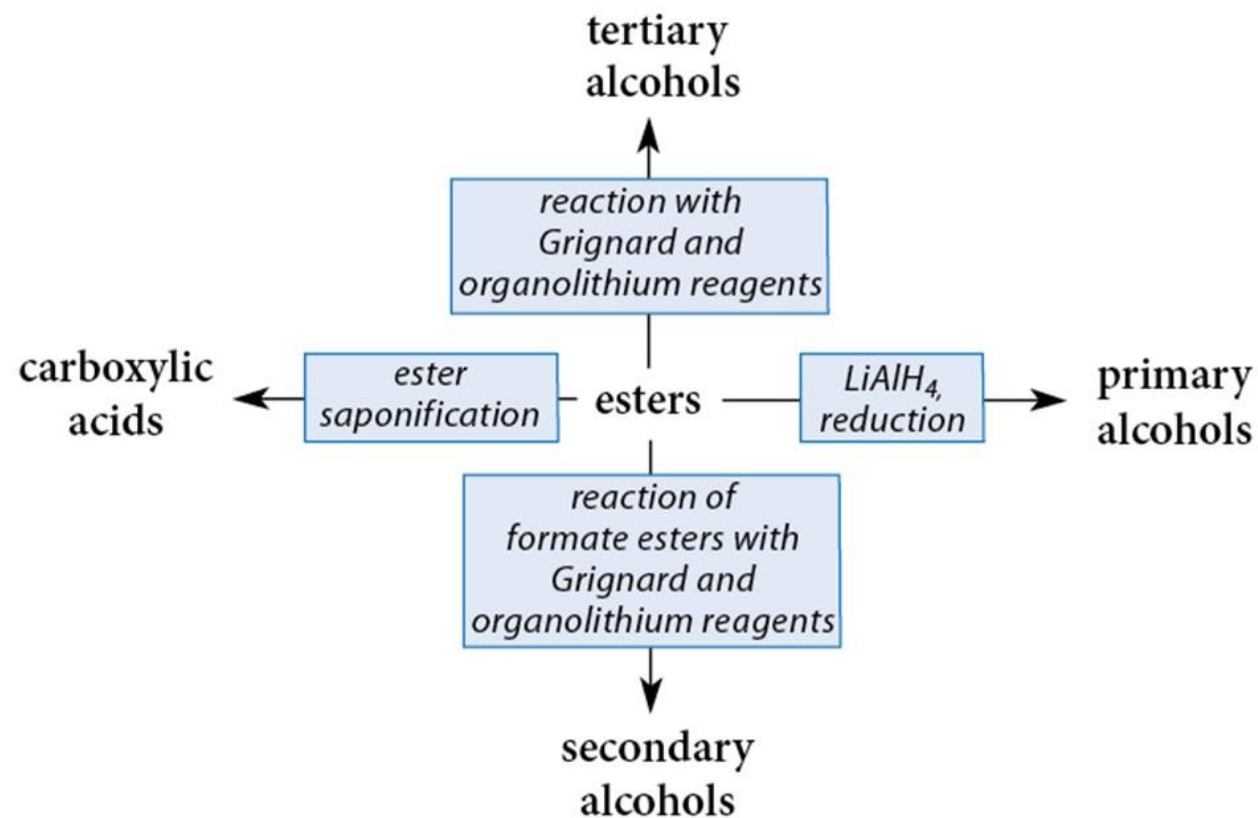
Synthesis of amides:

- Acid chlorides, anhydrides, or esters with amines

Synthesis of nitriles:

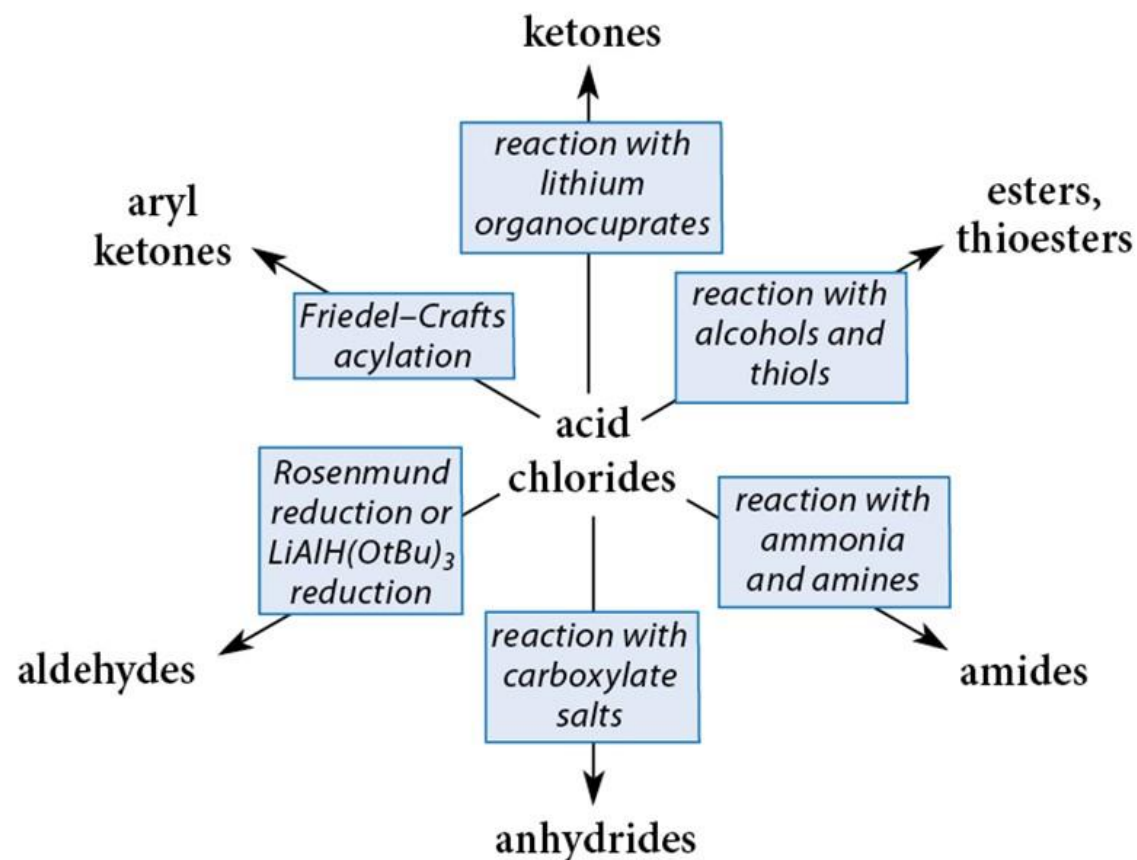
1. Cyanohydrin formation
2. $\text{S}_{\text{N}}2$ reaction of cyanide ion with alkyl halides or sulfonate esters

Esters in Organic Synthesis



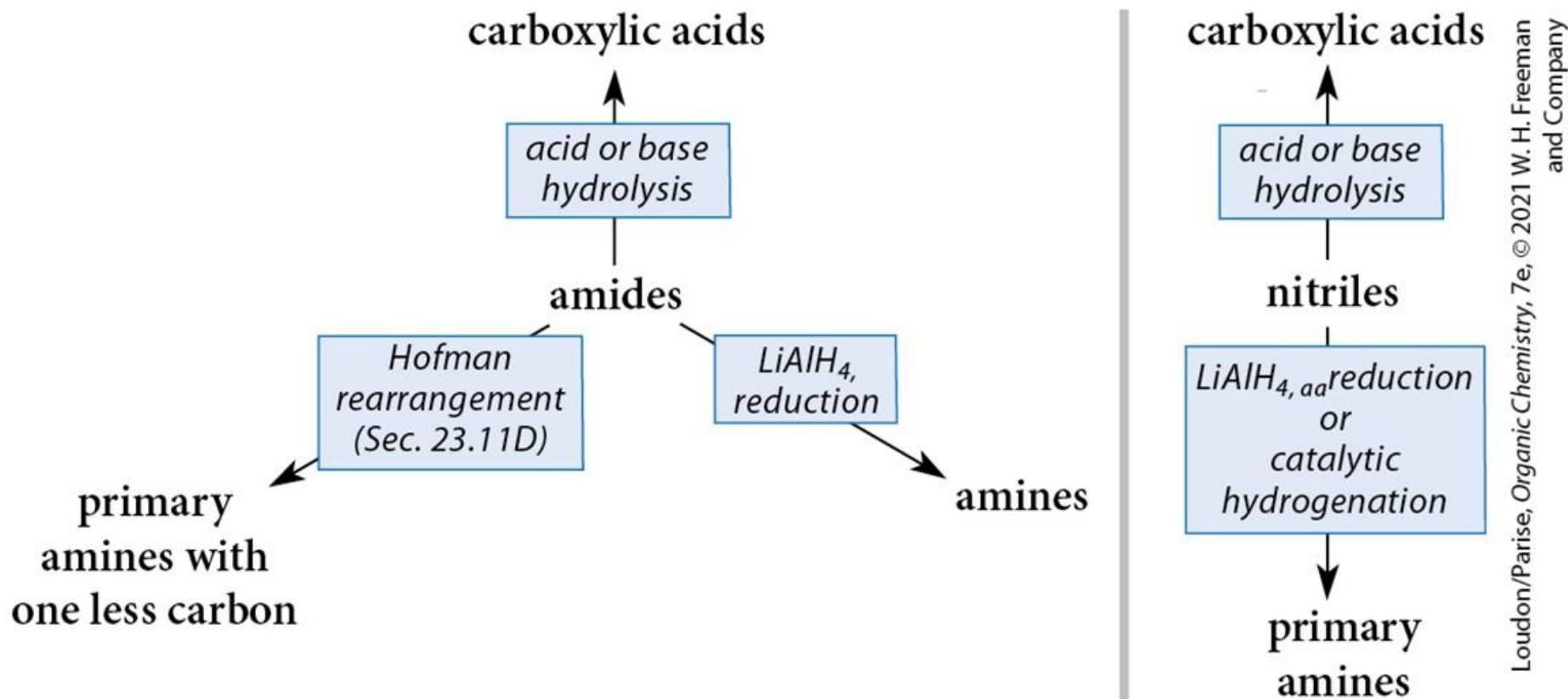
Loudon/Parise, *Organic Chemistry*, 7e, © 2021
W. H. Freeman and Company

Acid Chlorides in Organic Synthesis



Loudon/Parise, Organic Chemistry, 7e, © 2021 W. H. Freeman and Company

Amides and Nitriles in Organic Synthesis



Chapter 22 (Carey Chapter 21)

The Chemistry of Enolate Ions, Enols, and
 α,β -Unsaturated Carbonyl Compounds

Chapter 22 Overview

22.1 – Acidity of Carbonyl Compounds

22.2 – Enolization of Carbonyl Compounds

22.3 – α -Halogenation of Carbonyl Compounds

22.4 – Aldol Addition and Aldol Condensation

22.5 – Aldol Reactions in Biology

22.6 – Condensation Reactions Involving Ester Enolate Ions

22.7 – The Claisen Condensation in Biology: Biosynthesis of Fatty Acids

22.8 – Alkylation and Aldol Reactions of Ester Enolate Ions

22.9 – Conjugate-Addition Reactions

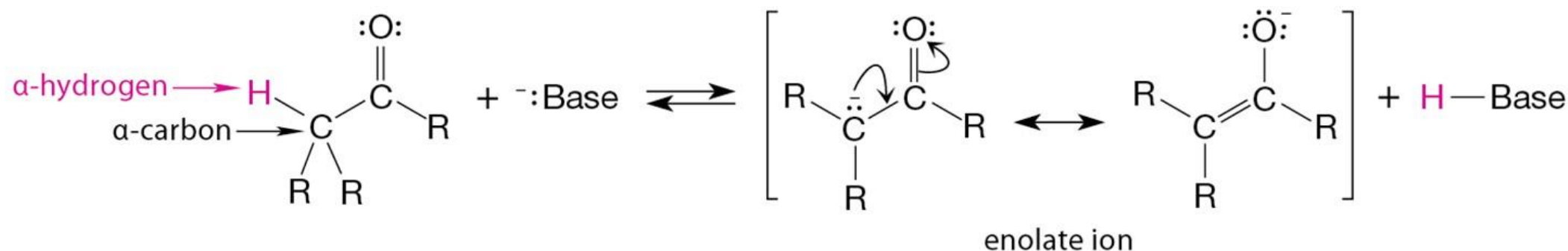
22.10 – Reduction of α,β -Unsaturated Carbonyl Compounds

22.11 – Reactions of α,β -Unsaturated Carbonyl Compounds with Organometallic Reagents

22.12 – Organic Synthesis with Conjugate-Addition Reactions

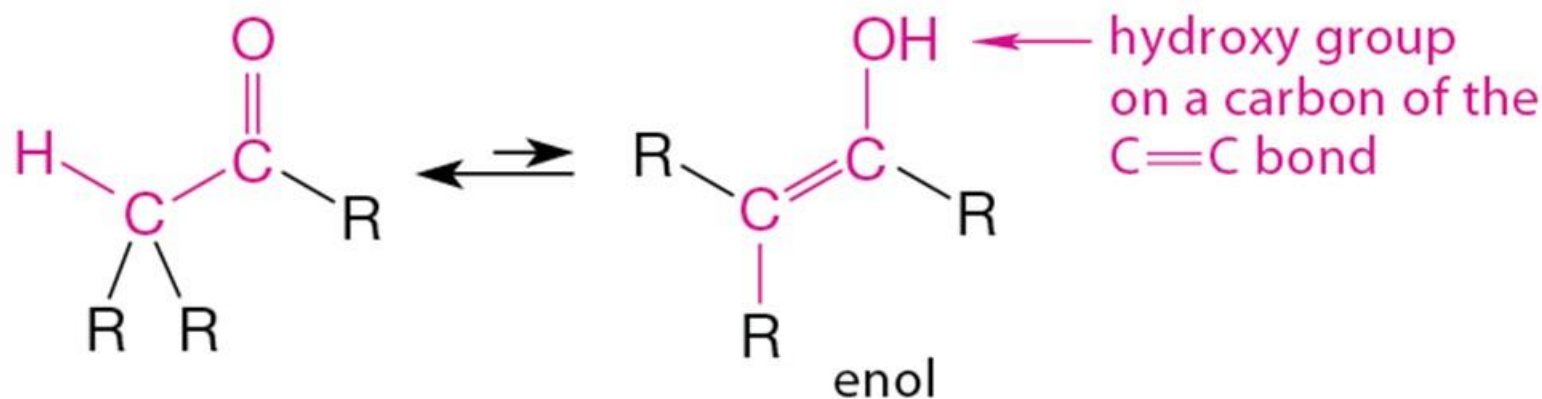
Enolates

- The α -H of carbonyl compounds are somewhat acidic.
- The conjugate base formed by the removal of an α -H is called an **enolation ion** or **enolate**.



Enols

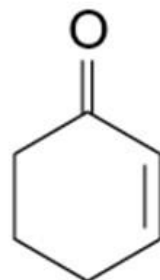
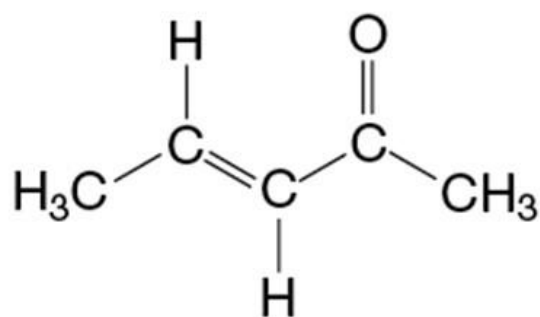
- An **enol** is any compound in which a hydroxy group is on a carbon of a C=C double bond.
- Most carbonyl compounds with α -H are in equilibrium with small amounts of their enol isomer.



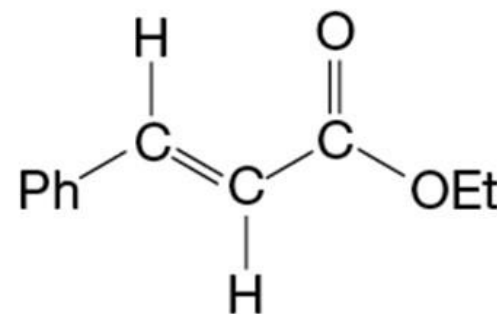
22.1 Acidity of Carbonyl Compounds

α,β -Unsaturated Carbonyl Compounds

- Compound in which a carbonyl group is conjugated with a carbon-carbon double bond.



α,β -unsaturated ketones
(sometimes called enones)

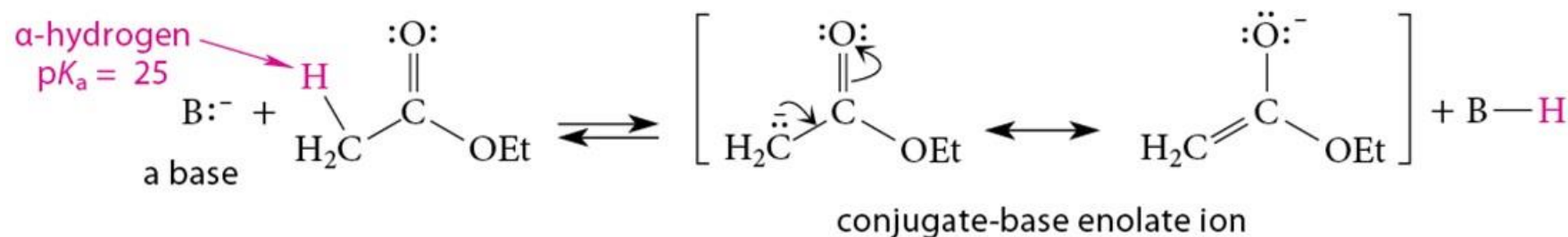
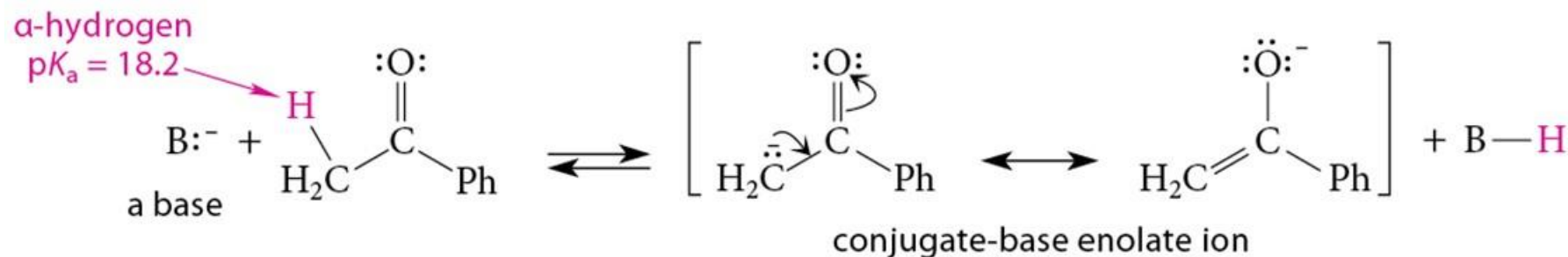


an α,β -unsaturated ester

22.1 Acidity of Carbonyl Compounds

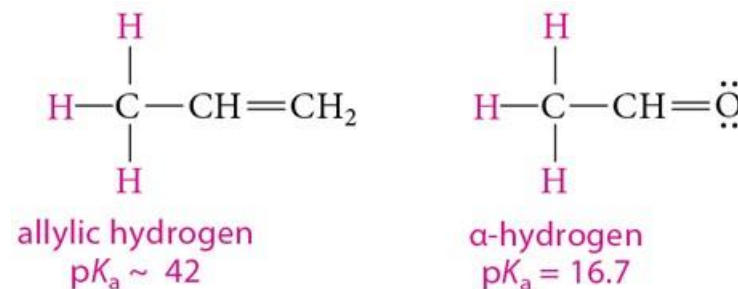
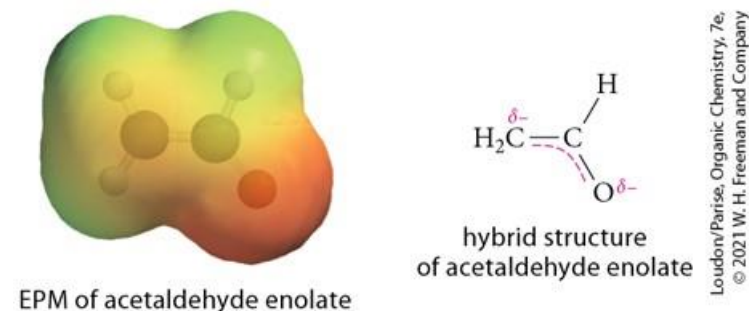
Formation of Enolate Anions: Ketones and Esters

- The α -H of aldehydes, ketones, esters, and even nitriles can be deprotonated.

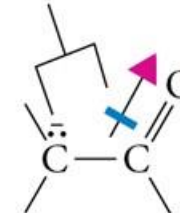


Acidity and Electron Distribution

- π_2 MO – α -carbon and carbonyl O are major sites of electron density.
- This is confirmed by EPM of enolate ion.
- The α -H of carbonyl compounds are more acidic than allylic H's of alkenes.

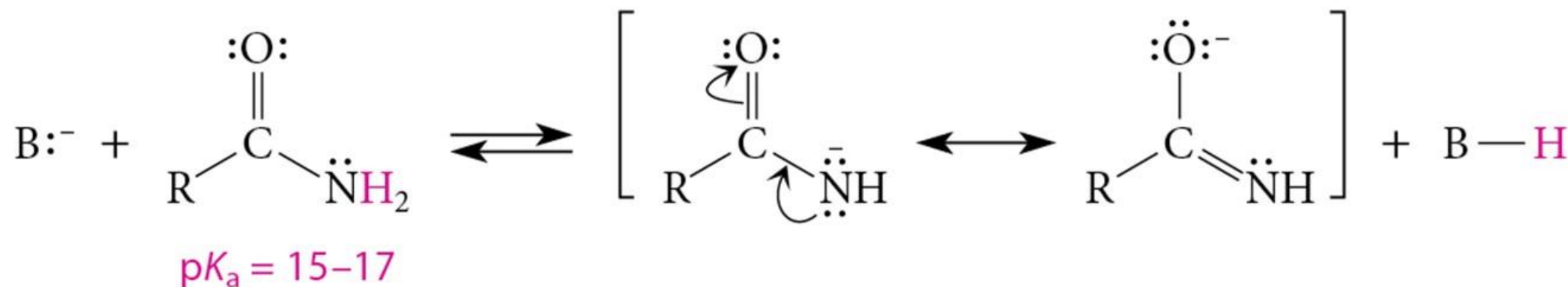


favorable charge-dipole interaction



Amide N—H Hydrogens Are α -Hydrogens

- The N—H of primary and secondary amides is more acidic than the α -H on an α -carbon.



Some multistep synthesis examples for your reference in the following slides.

COOH derivatives practice problems

- Multi-step synthesis 1

COOH derivatives practice problems

- Multi-step synthesis 1

COOH derivatives practice problems

- Multi-step synthesis 3 (this one can be tried out before the class if you want)

